

Sensors & Systems

Simple Sensors — Slides 1–9

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Slides 1–3 What Is a “Simple” Sensor?

A simple sensor is a self-contained device that converts a single, continuous physical quantity into an electrical signal that can be read directly with little or no additional computation. Three defining properties hold for all simple sensors:

- It measures one continuous physical quantity — temperature, magnetic field strength, sound pressure, humidity, etc.
- It provides a simple interface — either an analog voltage/current output or a straightforward digital protocol (I²C, SPI, UART, etc.).
- It does not require extensive processing to obtain the desired measurement — contrast this with a camera or radar, where significant computation must precede any useful output.

Examples encountered in this lecture include an air-quality sensor, a Hall effect sensor, and a microphone module — all pictured on slide 3. Despite measuring entirely different physical quantities, every one of them fits the same three-property mold.

figures/slide-03.jpg

Every sensor — simple or otherwise — follows the same general processing pipeline from the physical world to a number in your microcontroller. Understanding this chain is important because it tells you *where* a particular sensor sits and what processing remains your responsibility.



Figure 2: The sensor signal chain: physical event $\rightarrow$ sensing component $\rightarrow$ amplifier $\rightarrow$ ADC $\rightarrow$ computing.

Walking through each stage:

1. **Physical event.** A real-world phenomenon occurs — light, sound, temperature change, force, radiation. This is the stimulus the sensor exists to measure.

2. Sensing component. A transducer converts the physical phenomenon into an electrical signal, typically a voltage or current. This is the core physics step. The output is often tiny — millivolts or microamps.
3. Amplifier. The weak electrical signal is boosted to a level suitable for the ADC. Because sensor signals are small, amplifier quality matters; any noise or offset introduced here gets amplified alongside the signal.
4. Analog-to-Digital Converter (ADC). The amplified continuous signal is sampled at regular intervals T_s and converted to a digital integer. This is where the continuous physical world becomes a discrete-time sequence of values. Output protocols include PWM, RS232, I²C, SPI, and CAN.
5. Computing. A microcontroller or processor interprets the stream of digital values and acts on them.

Note — integration level. Different sensors integrate different numbers of these stages internally. A bare PT100 resistance thermometer gives you only a raw resistance — stages 2 onward are your problem. A TMP117 integrates the amplifier and ADC on-chip and hands you temperature in °C directly over I²C. Always check the datasheet to know what the sensor does for you.

Key Takeaway: The same five-stage chain governs every sensor system. “Simple” sensors merely make stages 3–4 (and sometimes 5) invisible to the user by integrating them on-chip.

section[]Slides 5–7 The Hall Effect Sensor

subsection[]What Is a Hall Sensor For?

A Hall sensor is used to measure or detect magnetic fields. Just as a thermistor converts temperature into a resistance that you can read, a Hall sensor converts magnetic field strength (flux density, measured in Gauss or Tesla) into a voltage that you can read.

Practical uses include:

- Detecting the presence of a magnet (e.g. reed-switch replacement, door/lid open-closed detection).
- Measuring rotation speed — a magnet on a shaft passes the sensor once per revolution, producing a pulse.
- Measuring current — a current-carrying conductor generates a magnetic field proportional to the current.
- Measuring the position of a magnetic actuator or joystick.

subsection[]The Physics of the Hall Effect (Slide 6)



Figure 3: Left: no magnetic field, $V_{\text{HALL}} \approx 0 \text{ V}$. Right: magnetic field perpendicular to bias current, V_{HALL} rises proportionally.

Inside the sensor there is a thin semiconductor sheet. Tracing through the signal chain of Figure ?? for this specific sensor:

Stage 1 — Physical event: A magnetic field is present (or not) in the vicinity of the sensor.

Stage 2 — Sensing component (the Hall element): A constant voltage source forces a fixed bias current I_{BIAS} through the semiconductor sheet along its length (the element marked with $\times$ in the left diagram).

- No field (left diagram): charge carriers flow straight through the sheet; the voltage measured *across the width* of the sheet, V_{HALL} , is approximately zero.
- Field present (right diagram): flux lines enter the semiconductor perpendicular to I_{BIAS} (the green arrows passing through the block). The Lorentz force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ deflects the moving charge carriers sideways, causing them to accumulate along one edge of the sheet. This charge separation creates a measurable transverse voltage — V_{HALL} rises toward a positive value.

The key relationship is:

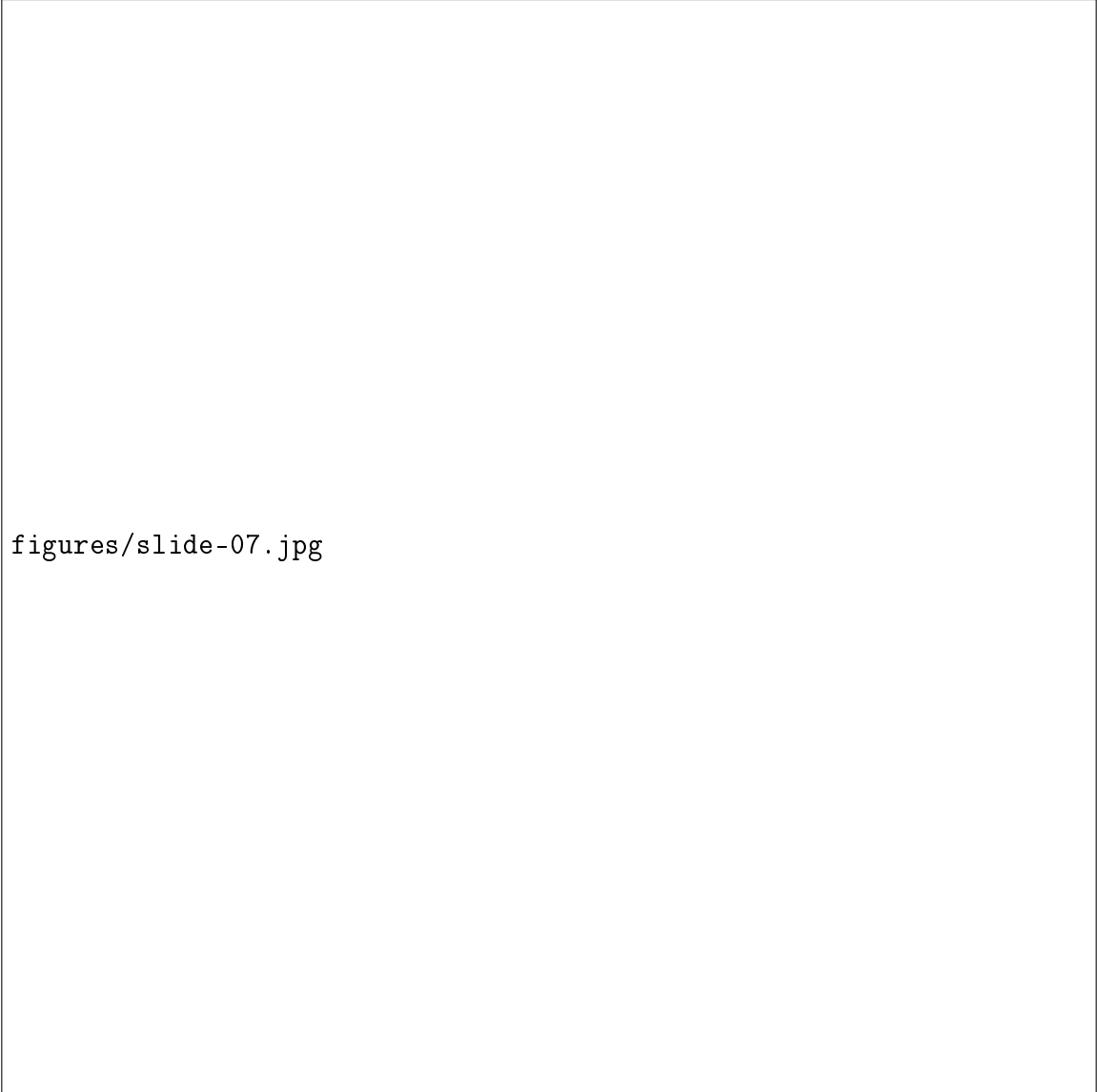
$$V_{\text{HALL}} \propto I_{\text{BIAS}} \cdot B$$

where B is the magnetic flux density. Since I_{BIAS} is kept constant by the circuit, V_{HALL} varies linearly with B — stronger field, larger voltage.

Variable summary for the Hall element:

- I_{BIAS} — constant bias current forced through the semiconductor sheet; set by the external voltage source.
- B — magnetic flux density of the external field (Gauss / Tesla); the quantity being measured.
- V_{HALL} — output voltage measured across the *width* of the sheet; approximately zero with no field, proportional to B when a field is present.

subsection[Signal Conditioning — Why Each Configuration Differs (Slide 7)]



figures/slide-07.jpg

Figure 4: Left: linear Hall sensor (Hall element + OpAmp only). Right: switch-type Hall sensor (Hall element + OpAmp + comparator/Schmitt trigger).

V_{HALL} is typically only a few millivolts — far too small to be useful on its own. This brings us to Stage 3 (Amplifier) in the signal chain. Two different goals lead to two different circuit configurations:

subsubsection[]Configuration 1 — Measuring Field Strength (left circuit)

If you need to know *how strong* the magnetic field is, you connect a precision operational amplifier (OpAmp) directly after the Hall element. The OpAmp amplifies V_{HALL} to a usable voltage level, and the analog output is proportional to B . In terms of the signal chain, you stop after Stage 3 (or pass to an ADC for Stage 4); you keep the output continuous and linear.

Warning: this configuration demands a high-quality, low-offset OpAmp. Any DC offset in the OpAmp is amplified just like the signal — a 1 mV offset with a gain of 500 becomes 500 mV of error, completely swamping a small magnetic signal.

subsubsection[[Configuration 2 — Detecting Presence Only (right circuit)

If you only need to know *whether* a magnet is nearby — a binary yes/no — you add a comparator (Schmitt trigger) after the amplifier. The comparator compares the amplified V_{HALL} against a threshold and outputs a clean digital HIGH or LOW.

Why is this circuit easier and cheaper?

- The comparator threshold absorbs small OpAmp offsets — the binary output is correct as long as the field is clearly above or below the threshold.
- A lower-grade OpAmp is acceptable because small gain errors do not affect the final digital output.
- The Schmitt trigger adds hysteresis: the turn-on and turn-off thresholds differ slightly, preventing rapid chattering when the magnet is near the boundary.

Key Takeaway — choosing a configuration:

- Need a *proportional* measurement of field strength? → Linear Hall sensor (OpAmp only). Requires a high-quality OpAmp; output is analog.
- Need only a *presence/absence* detection? → Switch-type Hall sensor (OpAmp + comparator). Tolerates lower-grade OpAmp; output is digital.

The two configurations differ because they have different goals — the stopping point in the signal chain (Stage 3 vs. a binary Stage 4) drives everything else.

section[[]Slide 8 The SS49E Linear Hall Sensor

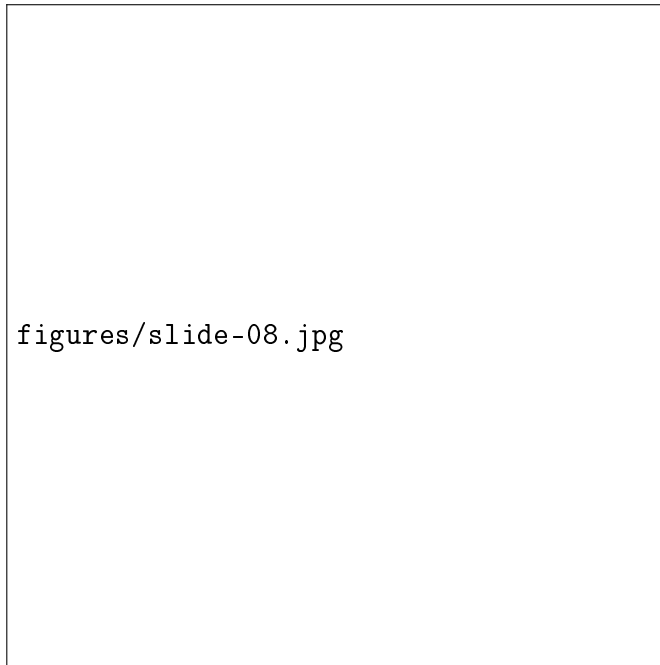


Figure 5: The SS49E linear Hall effect sensor module.

The SS49E is a specific linear Hall sensor — it uses Configuration 1 above and provides a continuous analog output proportional to field strength. Key specifications:

- **Sensitivity:** $2.5 \pm 0.5 \text{ mV/G}$. For every Gauss of magnetic flux density, the output voltage changes by approximately 2.5 mV. The $\pm 0.5 \text{ mV/G}$ tolerance means unit-to-unit sensitivity variation of up to 20% — calibration is advisable in precision applications.
- **Power consumption:** 4 mA at 5 V DC. Modest consumption; compatible with direct Arduino/microcontroller supply.
- **Minimum supply:** 2.3 V. The sensor can run from 3.3 V systems (Raspberry Pi, Nordic MCUs, etc.) without a level shifter.

Note — sensitivity tolerance. The $\pm 0.5 \text{ mV/G}$ spread means two sensors from the same batch may give readings that differ by up to 20% for the same field. For precise measurements, each unit should be individually calibrated against a known field.

section[Slide 9 Arduino Example — Reading the SS49E

figures/slide-09.jpg

Figure 6: Wiring diagram and Arduino code for reading the SS49E.

The example code on slide 9 demonstrates a complete reading pipeline. Three steps worth unpacking:

subsubsection[Step 1 — Averaging to reduce noise

```

long measure = 0;
for(int i = 0; i < 10; i++){
    measure += analogRead(pinHall);
}
measure /= 10;

```

Ten successive ADC readings are accumulated and then divided by 10. This is a basic averaging filter — a first encounter with a theme that runs throughout this lecture. Averaging N independent noise samples reduces the noise standard deviation by a factor of $\sqrt{N}$, so averaging 10 readings gives $\sqrt{10} \approx 3.2\times$ noise reduction at the cost of 10 ADC conversions.

subsubsection[Step 2 — ADC count to millivolts

```
float outputV = measure * 5000.0 / 1023;
```

The Arduino Uno’s ADC has 10-bit resolution: it maps 0–5 V onto the integer range 0–1023. Multiplying by 5000/1023 undoes this mapping and recovers the voltage in millivolts.

subsubsection[Step 3 — Millivolts to magnetic flux density

```
float magneticFlux = outputV * 53.33 - 133.3;
```

This is the inverse of the sensor’s sensitivity curve, converting millivolts back to flux density in millitesla. The coefficients 53.33 and 133.3 come from the SS49E datasheet. The subtraction of 133.3 accounts for the sensor’s quiescent output voltage at zero field (the output sits at mid-supply, not at 0 V, when no field is present — so zero field must be subtracted out).

Key Takeaways — Hall sensor end-to-end:

- A Hall sensor converts magnetic field strength to a voltage via the Hall effect.
- V_{HALL} is tiny; an OpAmp is always needed.
- For field *strength* measurements use a linear sensor (analog output). For field *presence* detection use a switch-type sensor (digital output).
- The SS49E has a sensitivity of $2.5 \pm 0.5 \text{ mV/G}$; calibration is recommended for precision work.

- Averaging multiple ADC readings is a simple and effective way to reduce measurement noise — a preview of the filtering concepts covered later in this lecture.